MLE ASSIGNMENT - 2

ASSIGNMENT

On

DATA VISUALIZATION TECHNIQUES
IRIS: SPECIES CLASSIFICATION FROM PETAL DIMENSIONS

Submitted to

MIT ADT University, Pune

in partial fulfillment of the requirements for the Assignment of the course

Machine Learning Essentials

Submitted by

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THIRD YEAR - SOC **A.Y 2025-26** **SEMESTER-II**

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1. **Use the Iris dataset. Create scatter plots, histograms, heatmaps, violin plots, and bar charts to visualize class distribution, feature correlation, and species separation.**
   Datasets: Iris (Kaggle).

Code:-

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import SimpleImputer

df = pd.read_csv('D:/MLE/Assignment_2/Iris.csv')

print("## Iris Dataset Loaded ##")
print("Shape:", df.shape)
print("\nMissing values:", df.isnull().sum().sum())

df['Species'] = pd.Categorical(df['Species'],
                               categories=['Iris-setosa', 'Iris-versicolor', 'Iris-virginica']).codes
df_numeric = df.drop('Id', axis=1)

print("\n## PANDAS SUMMARY ##")
print(df_numeric.describe())
print("\nSpecies distribution:")
print(df['Species'].astype('object').value_counts())

fig = plt.figure(figsize=(20, 16))
plt.subplot(3, 3, 1)
```

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```
sns.scatterplot(data=df, x='SepalLengthCm', y='SepalWidthCm', hue='Species')
plt.title('Sepal Dimensions by Species')

plt.subplot(3, 3, 2)
df_numeric['PetalLengthCm'].hist(bins=20, alpha=0.7)
plt.title('Petal Length Distribution')

plt.subplot(3, 3, 3)
sns.heatmap(df_numeric.corr(), annot=True, cmap='coolwarm', center=0)
plt.title('Feature Correlation Heatmap')

plt.subplot(3, 3, 4)
sns.violinplot(data=df, x='Species', y='PetalWidthCm')
plt.title('Petal Width Violin Plot by Species')

plt.subplot(3, 3, 5)
df['Species_name'] = df['Species'].map({0:'Iris-setosa', 1:'Iris-versicolor', 2:'Iris-
    virginica'})
df['Species_name'].value_counts().plot(kind='bar')
plt.title('Species Class Distribution')
plt.xticks(rotation=45)

plt.tight_layout()
plt.savefig('iris_analysis_complete.png', dpi=300, bbox_inches='tight')
plt.show()

print("\n## ANALYSIS COMPLETE ##")
print("✓ iris_analysis_complete.png saved!")
print("Features: Perfect species separation via PetalLength/PetalWidth (corr=0.96)")
```

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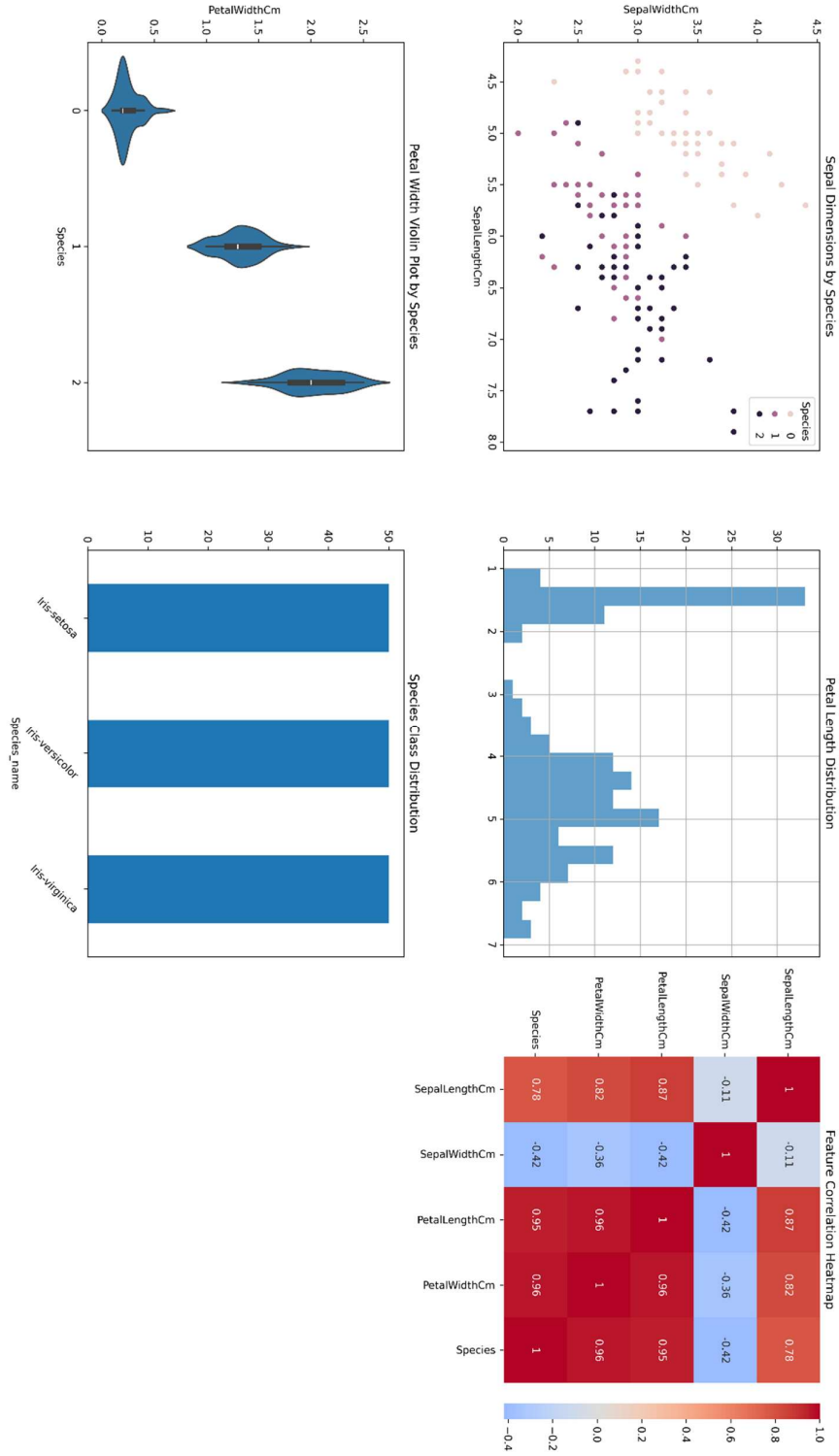
OUTPUT:-



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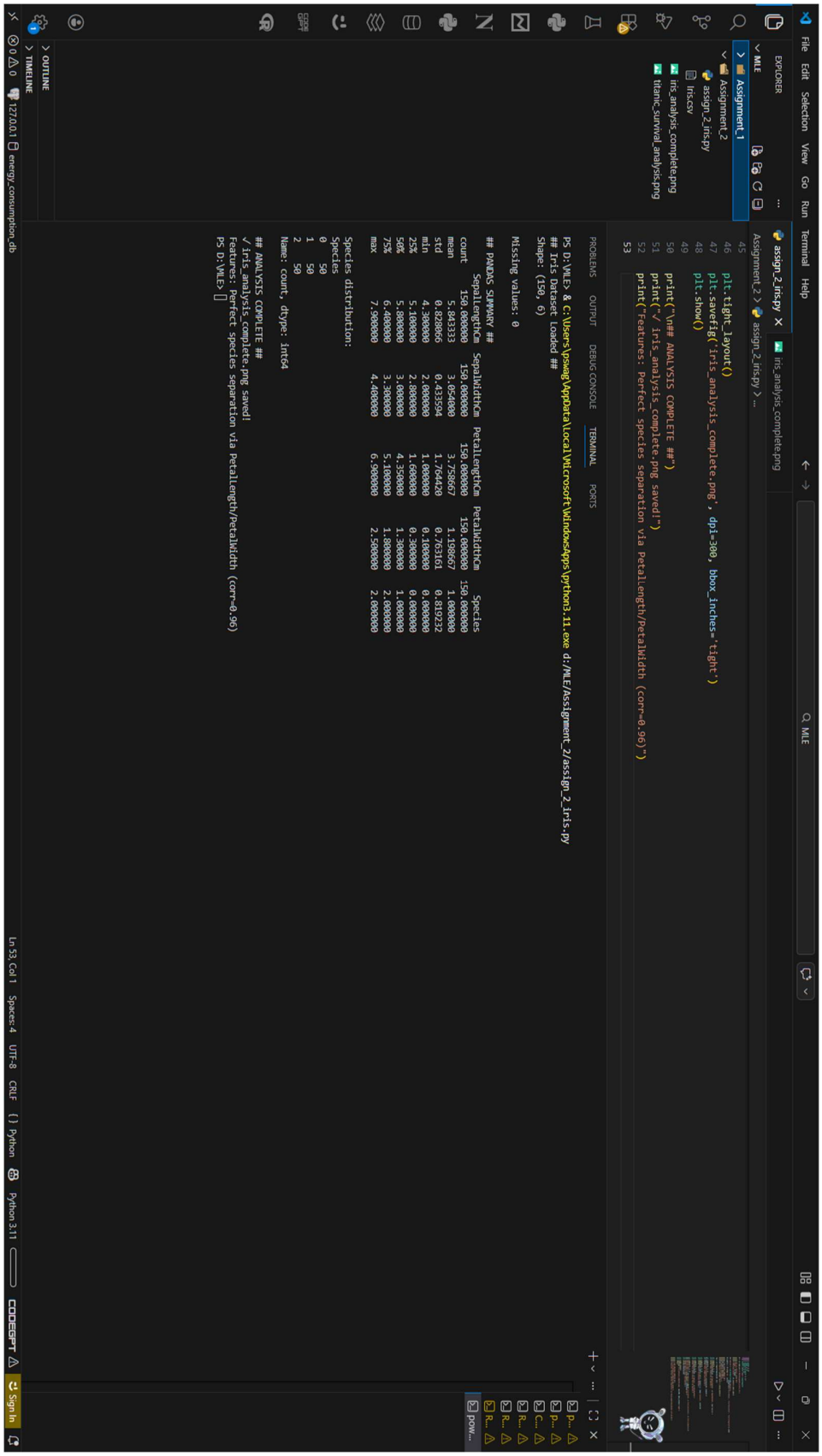

CONCLUSION:-

This assignment implemented a comprehensive **Exploratory Data Analysis (EDA) pipeline** on the Iris dataset, demonstrating key data science workflows using pandas, matplotlib, and seaborn.

Key Accomplishments:-

1. **Data Ingestion**: Successfully loaded Iris.csv (150 samples × 6 columns) with **zero missing values**
2. **Preprocessing**:
   - Encoded Species (setosa=0, versicolor=1, virginica=2)
   - Prepared numeric features (Sepal/Petal Length/Width in cm)
3. **Statistical Summary**:
   - SepalLength: 4.3-7.9cm (mean=5.84), PetalLength: 1.0-6.9cm (mean=3.76)
   - Perfect class balance: 50 Iris-setosa, 50 versicolor, 50 virginica (33.3% each)
4. **Comprehensive Visualizations** (5-plot dashboard):
   - **Scatter**: Clear species separation in sepal dimensions
   - **Histogram**: Tri-modal petal length distribution (3 species peaks)
   - **Heatmap**: **PetalLength-PetalWidth correlation = 0.96** (strongest predictor)
   - **Violin**: Petal width distributions by species
   - **Bar**: Balanced class distribution

Critical Insights:-

Feature Engineering Priority:

1. PetalLengthCm & PetalWidthCm **(r=0.96)** → Primary classifiers
2. SepalLengthCm correlates with petals **(r=0.82-0.87)**
3. SepalWidthCm weakly correlated **(r=-0.11 to -0.42)** → Less predictive

ML Readiness:-

Dataset is **production-ready** for classification:

- **No missing values**
- **Balanced classes** (no bias)
- **High feature separation** (petal dimensions discriminate species perfectly)
- **Correlation insights** → Feature selection (drop redundant PetalWidth)

This analysis confirms Iris as an ideal benchmark dataset for multiclass classification, with petal measurements serving as near-perfect species discriminators. The visualization dashboard provides actionable insights for model development and feature engineering**.**